

Coding & Solution

Date	30 June 2026
Team ID	
Project Name	A Comprehensive Measure Of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the implementation quality and completeness of the **Human Development Index (HDI) Prediction** project. The solution integrates a Machine Learning model with a Flask web application to predict the Human Development Index based on key socio-economic indicators. The project follows clean coding practices, modular design principles, and successfully satisfies the defined functional requirements.

Instructions:

1. The project code is organized into separate modules for data preprocessing, model training, prediction, and Flask web application development.
2. The solution addresses the Human Development Index prediction problem using a Machine Learning model and provides accurate prediction results through a user-friendly web interface.
3. All required source files, trained model files, HTML templates, datasets, and configuration files are included for project execution.
4. The project is maintained using Git and GitHub for version control and collaboration.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/kethineedilalitha09/Human_Development_Index
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	• Data preprocessing and feature engineering. • HDI prediction using a trained Linear Regression Machine Learning

	model. • User-friendly Flask web application. • Real-time prediction based on Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per Capita. • Prediction results displaying the estimated HDI score/category.
Pending / Incomplete Features	Cloud deployment, user authentication, database integration, prediction history storage, advanced data visualization dashboard.
Setup / Run Instructions	Install all required Python packages using requirements.txt, place the trained model file in the project directory, run app.py, and open the Flask server URL in a web browser.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- Successfully developed a Machine Learning-based Human Development Index Prediction System.
- Uses historical HDI data and socio-economic indicators for accurate prediction.
- Modular architecture and Flask deployment make the application easy to maintain and extend.
- Future enhancements include cloud deployment, interactive dashboards, user authentication, and support for additional machine learning models.